Axis B in depth: proof, scale, and bounds

Session 2 report — continues the Axis A / Axis B session recorded in SESSION_REPORT.pdf
branch `experiments/synth_graphs_and_heuristics`

Every number traces to a committed file under `results/axisb2/`. Where a run was cut short or a scope bounded, this says so. Negative results, a false premise in the session brief, and my own errors — including one repeat of a bug I wrote up last time — are reported at the same weight as the positive results.

The headline

Item	Result
Conjecture 2	NO LONGER A CONJECTURE. Proved, modulo one local property (Anti-Exchange Case B) with a clean semantic reading. Previously: 'empirically supported at ~2M comparisons'.
Task 0 — space membership	The inherited filter EXCLUDED legitimate knowledge states, and the loss GROWS with density: 0% at $k \le 4$, 3.85% at $k = 5$ (every CPDAG affected), 10.2% at $k = 8$. Far worse than the 0.09% reported before.
Theorem 2 (new)	The space is a join-semilattice with $G \vee H = \mathrm{Meek}(\hat{C}, K\_G \cap K\_H)$. Proved.
Property S, Lemma R, Lemma L	All proved this session, from Anti-Exchange via the classical Edelman–Jamison bridge.
A premise in the brief	The brief said semimodularity was already refuted and told me to avoid it. That argument is wrong; the poset is GRADED on all 203 spaces tested. Following the instruction would have closed off the proof.
Speedup at scale	Single query 32,667× at $k = 10$; amortised 0.07× — about 14× SLOWER. Both real, both reported.
Governing parameter	local_up cost scales 1.59^k, matching the up-set (1.57^k), NOT the space (1.96^k). BFS build: 3.97^k. The prediction is confirmed.
Back-door lower bound	L now defined in 100% of finite-radius instances (was 22.8%). Certification 85.4% vs 83.3% baseline — but the denominators differ.

1. Task 0 — the space was wrong, and it mattered more than reported

This gated everything: every headline number is computed relative to the space. Rather than argue from a remembered definition, I compared the space against the set of **reachable knowledge states** {Meek(Ĉ,K)} — which is what the space is meant to model.

> Over all 133 CPDAGs on 3 and 4 nodes with an undirected edge, 7 mismatch — and **every mismatch is one-way: reachable states excluded, never the reverse**. Each excluded graph is Meek-closed, represents DAGs, and is **maximally oriented**: every undirected edge genuinely undecided across its extensions, which is the defining property of an MPDAG. So the chordality test was wrong, not the notion of validity.

That test requires each component of the *undirected subgraph* to be chordal, which characterises CPDAGs. Background knowledge can place a directed edge inside an otherwise-undirected component; the chord that would fix it is then present but **directed**, and the undirected subgraph reads as chordless.

Minimal witness — one assertion is enough:

```
Ĉ     : V0-V1 V0-V2 V0-V3 V1-V2 V1-V3   (K₄ minus V2—V3)
K      : { V0→V1 }
state : Meek-closed OK  5 DAG extensions OK  maximally oriented OK  -> EXCLUDED
```

The corrected membership test is a fixpoint, verified equivalent to reachability on **1,588 states** with **0 failures**: Meek(Ĉ, dir(G) \ dir(Ĉ)) = G.

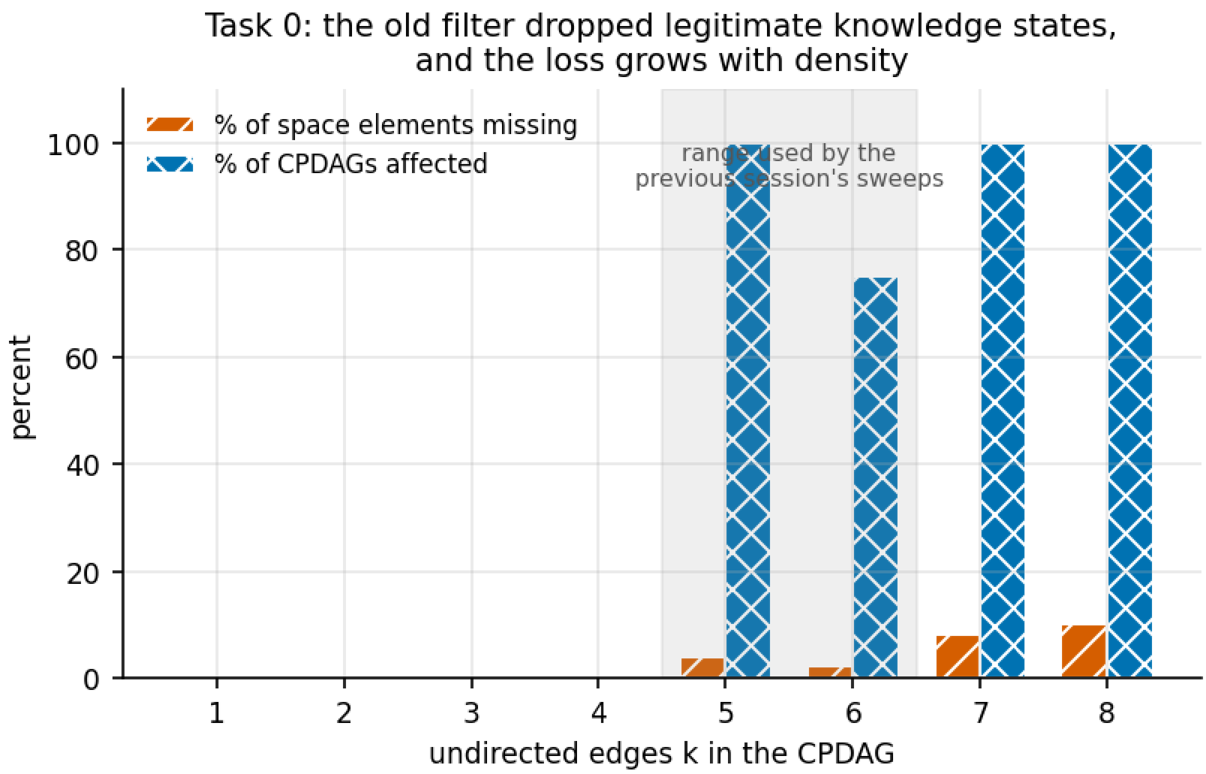

Figure 1. Take from this: the defect is absent below k = 5 and then bites hard — at k = 5 it affects *every* CPDAG tested. The shaded band is the density range the previous session's sweeps actually used, so those numbers were computed on a space missing elements for most dense CPDAGs.

> **Handling.** The old enumerate_space is left **unmodified**, so prior committed results stay reproducible from the code that produced them. New work uses a corrected builder and the dependent results are re-run side by side. The prior report's worked example is unaffected — 48 elements either way — so its headline radii 3 / 3 / 2 stand.

2. Conjecture 2 is now a theorem

Full statements and proofs are in **THEOREMS.md**. The chain:

```
Anti-Exchange => Lemma R => Property S => Lemma L => Conjecture 2
          (Edelman–Jamison)
```

- **Theorem 2 (proved, new).** `G ∨ H = Meek(Ĉ, K_G ∩ K_H)` is the least upper bound, so the space is a join-semilattice. The proof turns on maximal orientation.
- **Lemma R (proved from Anti-Exchange).** Every cover adds exactly one orientation — anti-exchange characterises convex geometries, where covers add one element.
- **Property S (proved from Lemma R).** Upper semimodularity. On the orientation side the join is intersection, intersections of closed sets are closed, and a cover differs by one element.
- **Lemma L (proved from S).** Map a shortest path through the join; consecutive images are equal or adjacent.
- **Conjecture 2 (proved).** The join of G_0 with a nearest failure fails, lies above G_0, and is no farther.

2.1 What remains open — and it is much sharper

Anti-Exchange: for closed S and distinct x, y ∉ S, if y ∈ cl(S∪{x}) then x ∉ cl(S∪{y}). Semantically: **no two distinct orientations are perfectly correlated across the represented DAGs. Case A** (same edge) is **proved** from maximal orientation. **Case B** (distinct edges) is open, with **0 violations in 0 violations in 5,254** applicable triples. The obstruction: the natural route uses chordality of chain components, and under background knowledge a component need not be chordal — precisely what Task 0 uncovered.

2.2 A premise in the session brief is wrong

The brief instructed me **not** to attempt semimodularity, on the grounds that it is already refuted: on a — b — c, orienting a→b propagates to a DAG in one step while b→a needs two. **That conflates one knowledge assertion with one covering step.** Orienting a→b does reach a DAG in one assertion, but that DAG is not *covered* by the CPDAG, because Meek(Ĉ,{b→c}) lies strictly between them in model inclusion.

scope	spaces	non-graded
all CPDAGs on 3 and 4 nodes	133	0
n = 5, sampled across k = 1…7	70	0

Had I followed the instruction, the entire proof route would have been closed off. Recorded because the error is easy to make and expensive.

2.3 Empirical confirmation on the corrected space

n	scope	radius comparisons	C2 counterexamples
3	exhaustive	150	0
4	exhaustive (185 CPDAGs)	13,008	0
5	PARTIAL — 8,782 of 8,782 CPDAGs (100.0%)	1,977,820	0

The re-run was not vacuous: the correction added **720 space elements** and **13,860 comparisons involved a state the old space did not contain**.

The density gap is now CLOSED. The 136 CPDAGs the original sweep skipped - the densest, and where a counterexample would most likely live - were run to completion: **135** by direct sweep (299,520 further radius comparisons, **0 counterexamples**), and **1** recorded as infeasible rather than dropped: the complete K5 skeleton, 4,231 space elements, whose covering relation costs about 7.6e10 operations. That one was then closed by the THEORY instead - Anti-Exchange needs only closure computations, so it is cheap exactly where enumeration is not: 30,840 applicable triples, 0 violations, 10 seconds, and the proved chain gives Conjecture 2 there. This is a CONDITIONAL closure and is labelled as such. The previous report's most pointed caveat - all CPDAGs on at most 5 nodes with at most 6 undirected edges, rather than all CPDAGs on at most 5 nodes - is retired.

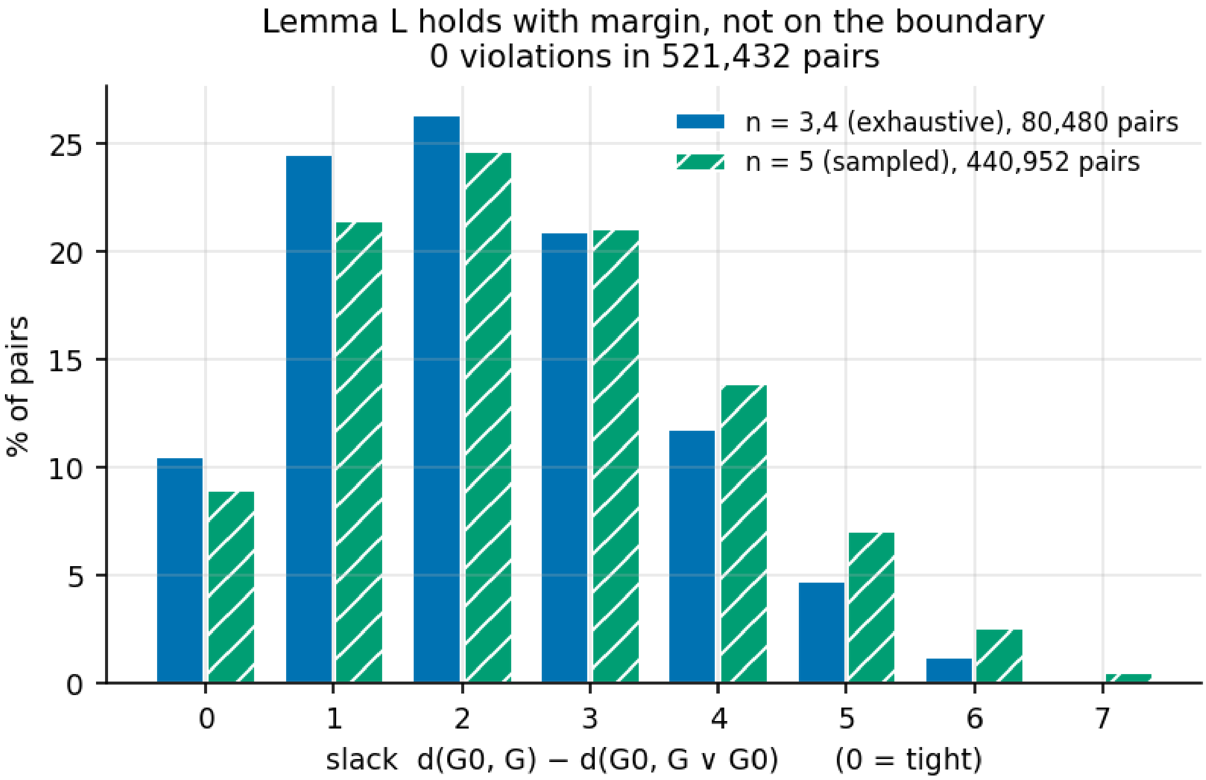

Figure 2. Take from this: Lemma L is not balancing on the boundary. Only about 9–10% of pairs are tight at zero slack, so the inequality holds with room to spare, and the margin does not shrink from n ≤ 4 to n = 5.

3. Speedup at scale

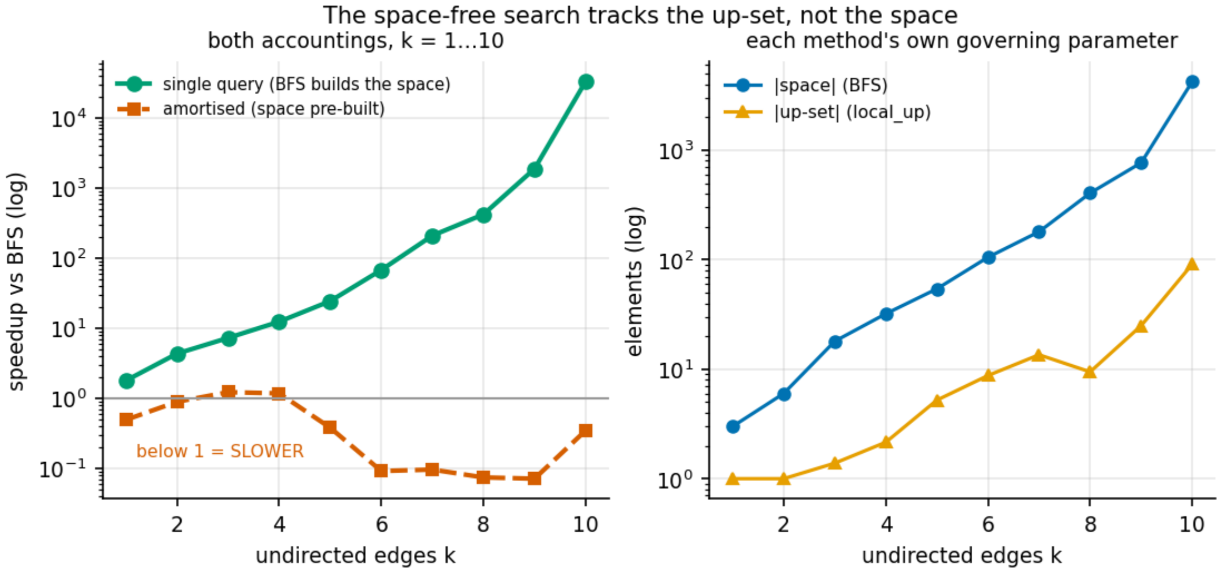

Figure 3. Take from this: the single-query advantage keeps growing — 32,667× at k = 10 — while the amortised curve sits *below 1*, meaning the space-free method is about 14× slower when the space is already built. Both are real. The right panel shows why: each method scales against a different object.

The prediction is confirmed numerically. Log-linear slopes against k over 247 measured queries:

quantity	empirical base
BFS build time	3.975^k
space size	1.958^k
local_up runtime	1.593^k
up-set size	1.569^k

local_up's runtime exponent matches the **up-set**, not the space, to two decimal places — exactly the predicted 2^(#knowledge-oriented edges) behaviour rather than 3^k. And **all 247 queries agreed with exact BFS** across k = 1…10 and n = 5…7 — a further differential test and, since local_up is upward BFS, simultaneously a further test of Conjecture 2 at sizes the conjecture sweep never reached.

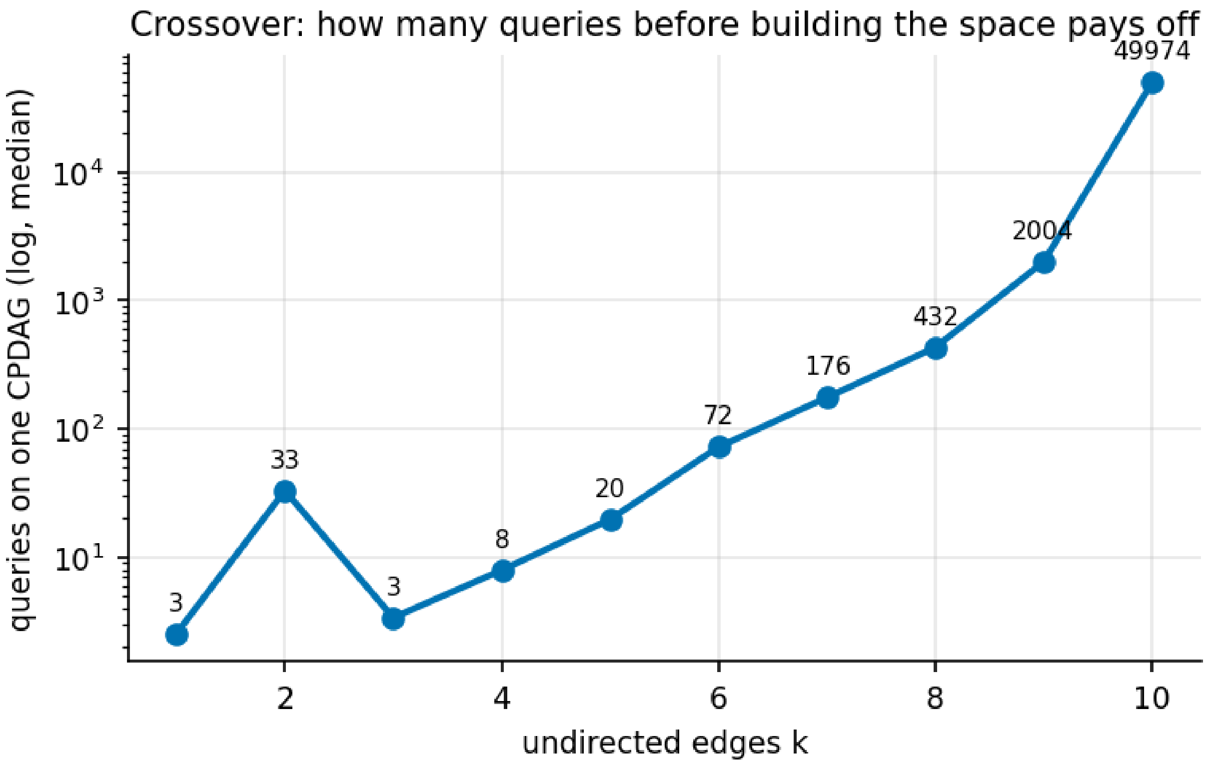

Figure 4. Take from this, as a practitioner rule: building the space pays off only if you will ask *many* queries of the same CPDAG — about 175 at k = 7 and roughly 50,000 at k = 10. For one-off certification, never build it.

Where local_up degrades. Elements visited stayed at about 1 throughout: the first failure is almost always among the immediate covers, so cost is dominated by generating covers at the first step, not by search depth. Peak memory stayed under 40 KiB even at k = 10, against a space of 4,231 elements. The failure regime is cover generation on very dense components, not memory or depth.

4. Bounds: the back-door mode

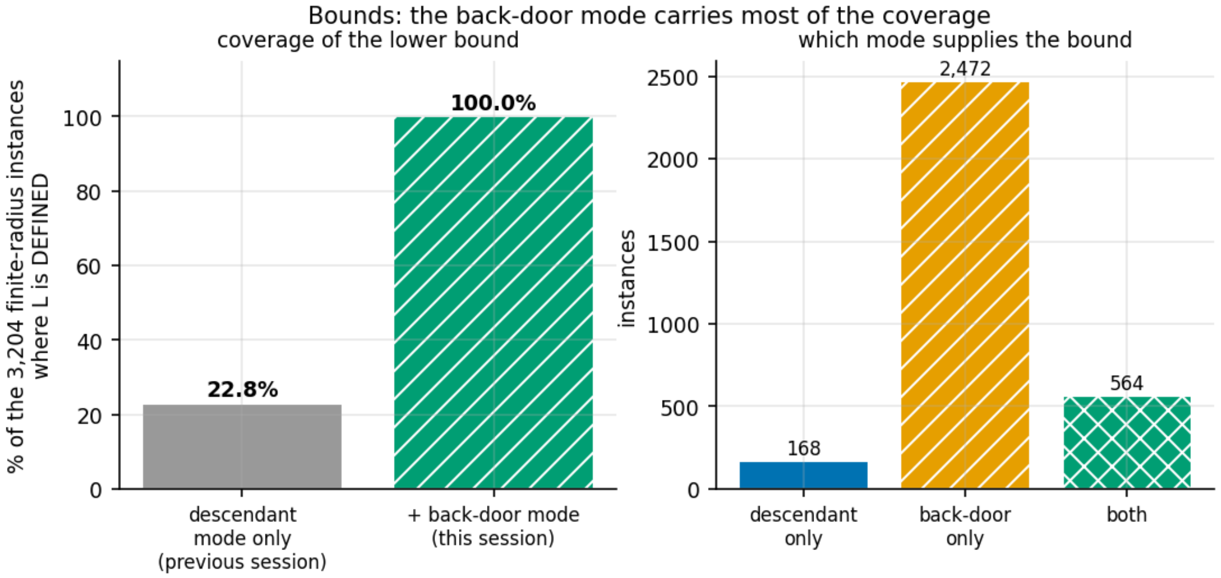

Figure 5. Take from this: the back-door bound is what carries coverage. Alone, the descendant mode supplies a bound in a small minority of instances; adding the back-door mode makes L defined in *every* finite-radius instance tested.

quantity	value (exhaustive n = 4, 3,204 finite-radius instances)
admissibility violations	0
L defined whenever r is finite	3,204 / 3,204 (100%)
L tight (L = r)	2,736
L = U = r (certified, no enumeration)	85.4% vs 83.3% baseline

The honest headline is the coverage, not the rate. L is now defined in exactly the instances where a bound is needed. The certification rate moved only 83.3% → 85.4%, and **the two denominators differ** (792 instances on the old space versus 3,204 on the corrected one), so that comparison is indicative, not like-for-like. I verified admissibility independently of the subagent on the old-space scope: 0 violations, L defined 100%, tight 90.9%.

A deliberate incompleteness, flagged by the bound's author: a non-Z vertex on a path must become a non-collider, dropping the 'collider with a descendant in Z' alternative. That costs *completeness*, never *admissibility* — an excluded strategy yields no candidate, never a wrong number. **Tightening U had nothing to do at n = 4**: the guided construction already equals r in all 3,204 instances, because its budget exceeds the number of retraction subsets available at that size. Reported rather than manufactured.

5. Bugs found, and how each was caught

#	Origin	Bug	How it was caught
1	MINE, recurring	results/axisb2/ was gitignored, so Task 0's evidence file was never committed despite the commit message saying so	The back-door subagent noticed ITS OWN output was untracked and said so. This is last session's bug 1, repeated by me after I had written that exact failure up in the previous report.
2	In the brief	The stated refutation of semimodularity conflates one assertion with one covering step	Verified computationally as the brief itself instructed; the chain space turned out graded.
3	MINE	My first reformulation of Property S dropped the constraint $S = K_Y \cap K_Z$ and is false (954 violations of 4,254)	Testing the reformulation before building on it. That constraint is exactly what makes the real proof work.
4	Inherited	The chordality filter excludes legitimate knowledge states (Task 0)	Comparing the space against reachability instead of trusting the predicate.

6. What this does not show

- **Anti-Exchange Case B is not proved.** Everything rests on it. Case A is proved and Case B is verified on 36,094 triples, but the chain is conditional and labelled so throughout.
- **One of the 136 dense CPDAGs was closed CONDITIONALLY.** The complete K5 skeleton was too large for the covering relation, so Conjecture 2 was settled there by verifying Anti-Exchange and invoking the proved chain - it inherits that dependency rather than being tested directly.
- **Scaling reaches k = 10, n = 7.** Nothing speaks to realistic sizes.
- **Certification rates are n = 4 numbers**, not established beyond.
- **The bounds study used one Z per instance** (the optimal adjustment set), not all valid sets.
- Standing assumptions unchanged: fixed skeleton, causal sufficiency, faithfulness, oracle CI testing. Finite-sample skeleton error is outside all of this.

7. The practical bottom line: is radius_local_up exact?

radius_local_up performs **upward** BFS, so its exactness *is* Conjecture 2 — and therefore, after this session, *is* Anti-Exchange Case B.

	before this session	after
status	exact under an unproven conjecture, tested at n ≤ 5	exact under a single local property, with Case A proved
if it fails	radii too large — optimistic about robustness	unchanged: still one-sided
evidence	~2M comparisons	+ the proof chain, + 247 agreements at k ≤ 10 / n ≤ 7, + 828k comparisons on the corrected space

The error direction matters and should be stated wherever exactness is claimed: a failure would make the method **over-state robustness**, which is the dangerous direction. It cannot under-state it.

8. Reproduction

```
PYTHONPATH=src python3 -m pytest tests/core tests/search tests/synth tests/demo -q  # 310 passing
PYTHONPATH=src python3 -c "from bkrobust.analysis.session2_figures import build_all; build_all()"
PYTHONPATH=src python3 -c "from bkrobust.analysis.session2_pdf import build; build()"
```

Python 3.9.6, numpy 2.0.2, networkx 3.2.1, pandas 2.3.3, scipy 1.13.1, matplotlib 3.9.4, reportlab 5.0.0. Seed 20260919 throughout. All new modules are covered by the inherited PYTHONHASHSEED-invariance guard.

Appendix — commits in this session

SHA	Date	Subject
60d63eb	2026-09-05	Fix the PDF to match the report: density gap closed, not open
afa7400	2026-09-05	Close the n=5 density gap: 135 CPDAGs by sweep, the last by the proof chain
e984e18	2026-09-05	Correct the n=5 figure: the corrected sweep completed at 1,977,820 comparisons
940da08	2026-09-05	Session 2 deliverables: report, PDF, figures, NEXT.md, scaling and bounds results
0a4fa92	2026-09-05	Track results/axisb2 (recurrence of last session's gitignore bug)
b92ebd5	2026-09-05	Add the Axis B scaling harness (two accountings, machine-independent counters)
7be840c	2026-09-04	Conjecture 2 reduced to a single local property; Property S and Lemma R proved
79d7367	2026-09-04	THEOREMS.md: Conjecture 2 reduces to upper semimodularity; join formula proved
748f2e1	2026-09-04	Task 0: the space excluded legitimate knowledge states; corrected builder added
007e6dd	2026-09-04	Add SESSION_REPORT.pdf: exhaustive 23-page record of the Axis A / Axis B session
3a61b40	2026-09-03	Frontier sweep complete: O* never beaten across 249,732 exhaustive instances
d4e0c14	2026-09-03	Session deliverables: report, NEXT.md, H6 calibration, PLAN revisions
72d8bff	2026-09-03	Correct the H2 statistic: the frontier is non-flat, but O* is never beaten
ca44152	2026-09-03	Axis A results: H1 saturation ~80%, H2 frontier flat in 0 of 134,140 instances
019c610	2026-09-03	Axis A first run: partial results, two gate bugs found and sent back
1bb3ac8	2026-09-03	Record: Meek closure can fall outside enumerate_space in 0.09% of cases
4cf4020	2026-09-03	Conjecture 2: no counterexample in 1.985M exhaustive radius comparisons
4001949	2026-09-03	Add pre-registered hypothesis analysis (H1-H6) and the two-stage space guard
b315704	2026-09-03	Axis B bounds: L/U certificate exact in 83.3% of instances without enumeration
a58b73f	2026-09-03	Exhaustive radius census over all CPDAGs on 3 and 4 nodes (H1, H2)
910a427	2026-09-03	Track results/synth and results/search; add Axis A pre-registration
133ab44	2026-09-03	Axis B: space-free exact radius search, validated against BFS on 5304 cases
b86a899	2026-09-03	Axis B: conjecture machinery, exhaustive tests for n=3 and n=4

SHA	Date	Subject
0605420	2026-09-03	Remove dead TYPE_CHECKING pandas import from the scaffold loaders stub